

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A SHORT ANSWER TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I B. Bhargavi 192311136 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

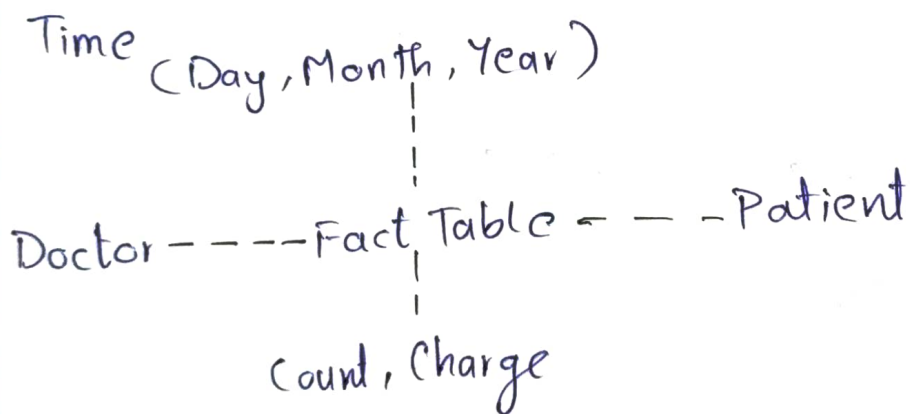
Data Warehouse : Doctor, Patient, Time

a) Three popular schemas

The three commonly used schemas in a data warehouse are:

1. **Star Schema** - One fact table connected to dimension tables. It is simple and fast.
2. **Snowflake Schema** - Dimension tables are further divided into smaller tables to reduce duplication.
3. **Fact Constellation (Galaxy) Schema** - Contains multiple fact tables sharing common dimensions.

b) Star Scheme Diagram



Fact Table

- * Count (Number of visits)
- * Charge (Doctor's fee)

Dimension Tables

- * Time (Day, Month, Year)
- * Doctor (Doctor Name, Hospital.)
- * Patient (Patient Name, Age)

c) SQL Query

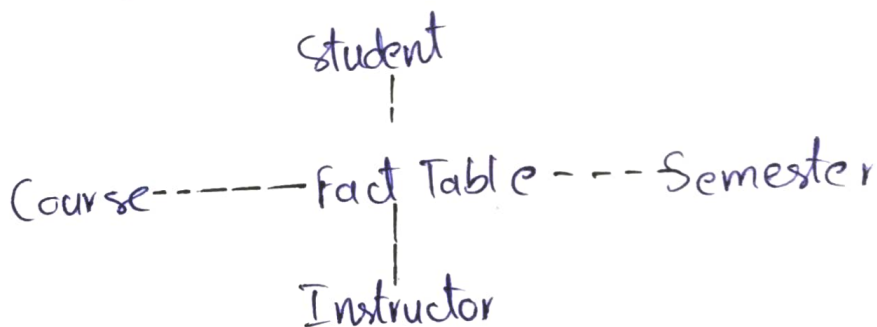
```
SELECT year, month, doctor,  
SUM(count) AS Total-Visits,  
SUM(charge) AS Total-Charge  
FROM fee  
GROUP BY year, month, doctor;
```

Explanation:

This query shows the total number of patient visits and total fees collected by each doctor every month.

2 Big University Data Warehouse

a) Snowflake Schema



Fact Table

- * Count
- * Average Grade

Dimensions

- * Student
- * Course
- * Semester
- * Instructor

b) OLAP Operations

To find the average grade of CS courses for each student!

1. Roll-up Semester to Year.
2. Roll-up Instructor to All.
3. Slice Course = CS
4. Calculate the average grade for each student.

c) Number of Cuboids

Formula:

$$5^4 = 625$$

Answer: 625 cuboids

3. Design a Weather Data Warehouse

The weather bureau collects:

- * Temperature
- * Air Pressure
- * Rainfall (Precipitation)

every hour from 1,000 weather probes.

Fact Table

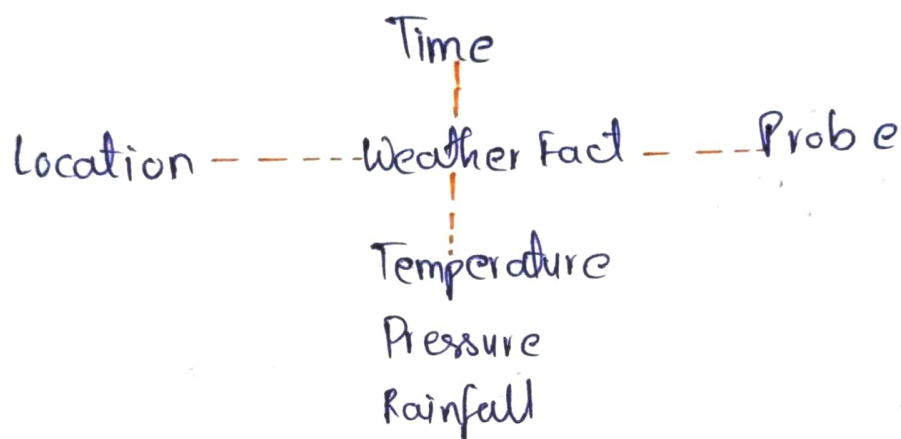
- * Temperature
- * Air Pressure
- * Precipitation

Dimension Tables

- * Time: Hour, Day, Month, Year
- * Location: City, Region, Land / Ocean

* **Probe**: Probe ID, Probe Type

Simple Diagram



Uses

- * Find average temperature by month.
- * Compare rainfall in different regions.
- * Study weather changes over several years.

4. Data Cube

Given

n = Number of dimensions

P = Number of values in each dimension

a) Maximum number of cells in the base cuboid

$$\boxed{P^n} \text{ Base Cuboid(Max)}$$

b) Minimum number of cells in the base cuboid

$$\boxed{1} \text{ Base Cuboid(Min)}$$

c) Maximum number of cells in the complete data cube

$$\text{Complete Cube(Max)} = (P+1)^n$$

d) Minimum number of cells in the complete data cube

$$\text{Complete Cube(Min)} : 2^n$$